SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 5 **Duration : 1 Hour**

Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I K.Niharika(192411124) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

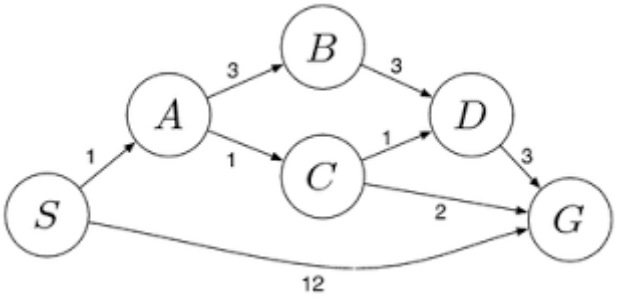
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

SOLUTIONS

Question 1 :Water Jug Problem

Problem: We are given a 4-gallon jug and a 3-gallon jug. Neither has a measuring scale. Using a water pump and the allowed operations, obtain exactly 2 gallons in the 4-gallon jug.

1.1 State Representation

Represent a state as (x, y), where x is the amount of water in the 4-gallon jug and y is the amount in the 3-gallon jug.

Initial state: (0, 0)

Goal state: (2, 0)

1.2 Allowed Actions

- Fill the 4-gallon jug.
- Fill the 3-gallon jug.
- Empty the 4-gallon jug.
- Empty the 3-gallon jug.
- Pour water from the 4-gallon jug into the 3-gallon jug.
- Pour water from the 3-gallon jug into the 4-gallon jug.

1.3 Step-by-Step Solution

Step	Action	4-Gallon Jug	3-Gallon Jug

1	Start	0	0
2	Fill 3-gallon jug	0	3
3	Pour 3-gallon into 4-gallon	3	0
4	Fill 3-gallon jug again	3	3
5	Pour 3-gallon into 4-gallon until full	4	2
6	Empty the 4-gallon jug	0	2
7	Pour the remaining 2 gallons into 4-gallon jug	2	0

Final state: (2, 0)

Conclusion: Exactly 2 gallons are obtained in the 4-gallon jug.

Question 2 : Mars Rover as an Intelligent Agent

Problem: Analyze how a Mars Rover explores Mars and collects scientific samples.

2.1 Percepts

Percepts are the information received from the environment through sensors.

- Camera images and visual information.
- Distance and obstacle information.
- Temperature and pressure readings.
- Soil and rock information.
- Rover position and orientation.
- Sample and terrain information.

2.2 Operating Environment

Property	Mars Rover Environment	Reason
Observability	Partially observable	The rover cannot observe the entire Mars surface.

Determinism	Stochastic	Sensor readings and movement can have uncertainty.
Episodes	Sequential	Current actions affect future decisions.
Change	Dynamic	Environmental conditions may change.
State	Continuous	Position, speed, temperature etc. can vary continuously.
Agents	Single-agent	The rover mainly operates independently.

2.3 Actions

- Move forward or backward.
- Turn left or right.
- Stop.
- Avoid obstacles.
- Collect soil or rock samples.
- Analyze samples.
- Take photographs.
- Send scientific data to Earth.

2.4 Performance Measure:

The rover's performance can be evaluated using:

- Amount and quality of useful scientific data collected.
- Number and quality of samples collected.
- Accuracy and safety of navigation.
- Distance explored.
- Energy consumed.

- Time taken.
- Successful communication with Earth.

Overall performance: Maximum useful exploration and scientific information with minimum risk, time and energy.

2.5 Suitable Agent Architecture

Answer: Utility-Based Agent

A utility-based agent is suitable because the rover must balance several factors such as scientific value, safety, energy consumption and time. It can select the action that gives the best overall utility.

Question 3: 8-Queens Problem

Problem: Place 8 queens on an 8 × 8 chessboard so that no two queens attack each other.

3.1 State Representation

Represent each queen by its (row, column) position. The required solution is:

Q1 → (1,4)

Q2 → (2,6)

Q3 → (3,8)

Q4 → (4,2)

Q5 → (5,7)

Q6 → (6,1)

Q7 → (7,3)

Q8 → (8,5)

State representation: [4, 6, 8, 2, 7, 1, 3, 5]

3.2 Initial State

Initially, the 8 × 8 board is empty. Queens are placed row by row and every new position is checked for conflicts.

3.3 Goal Test

The goal is reached when all 8 queens are placed and no two queens attack each other.

No two queens are in the same row.

No two queens are in the same column.

No two queens are on the same diagonal.

Diagonal condition: For queens (r1,c1) and (r2,c2), a diagonal conflict occurs when $|r1 - r2| = |c1 - c2|$.

3.4 Step-by-Step Placement

Step	Queen	Position	Check
1	Q1	(1,4)	No conflict
2	Q2	(2,6)	No conflict with Q1
3	Q3	(3,8)	No conflict with Q1, Q2
4	Q4	(4,2)	No conflict with Q1–Q3
5	Q5	(5,7)	No conflict with Q1–Q4
6	Q6	(6,1)	No conflict with Q1–Q5
7	Q7	(7,3)	No conflict with Q1–Q6
8	Q8	(8,5)	No conflict with Q1–Q7

3.5 Final Board

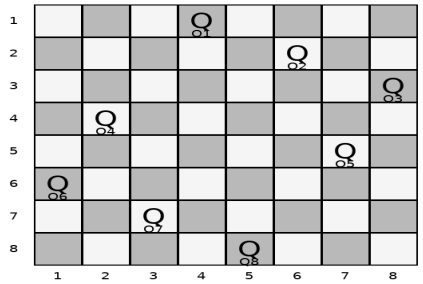


Figure: Final valid 8-Queens configuration

Final state: [4, 6, 8, 2, 7, 1, 3, 5]

Conclusion: All 8 queens are placed successfully with no row, column or diagonal conflicts.

Question 4 :OLA Cab Booking – Problem-Solving Agent

Problem: A customer wants to travel from one location to another using an OLA Cab booking application. The customer can choose Mini, Micro, Sedan, Shared or Prime.

4.1 Suitable Agent

Answer: Goal-Based Agent

A goal-based agent is suitable because the main goal is to take the customer from the source location to the destination. The agent selects actions that help achieve this goal.

4.2 Problem Formulation

Component	Description
Initial State	Customer is at the source location.
Goal State	Customer reaches the destination.
Actions	Select cab, search availability, book cab, assign driver and navigate.
Constraints	Cab must be available and the selected cab type must be available.

4.3 Pseudocode

START
Input source
Input destination
Input cab_type
IF source == destination
Display "Already at destination"
STOP
END IF
Search for available cab of cab_type

IF cab is available

Show fare and estimated time

Ask customer for confirmation

IF customer confirms

Book cab

Assign driver

Navigate to destination

Display "Reached destination"

ELSE

Display "Booking cancelled"

END IF

ELSE

Display "Cab not available"

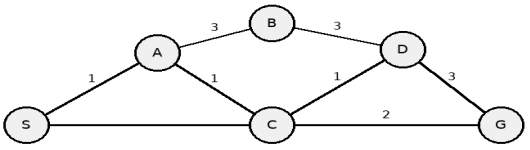
END IF

STOP

Question 5:Uniform Cost Search (UCS) – Logistics Delivery Network

Problem: Find the least-cost path from starting warehouse S to destination warehouse G using Uniform Cost Search.

5.1 Given Weighted Graph



5.2 UCS Principle

Uniform Cost Search always expands the node with the lowest total path cost g(n) from the start node S.

Rule: Choose the lowest cumulative cost first.

5.3 Step-by-Step Search

Step	Expanded Node	Path	Cost
1	S	S	0
2	A	S → A	1
3	C	S → A → C	2
4	D	S → A → C → D	3
5	B	S → A → B	4
6	G	S → A → C → G	4

5.4 Cost Calculation

S → A → C → G = 1 + 1 + 2 = 4

S → A → C → D → G = 1 + 1 + 1 + 3 = 6

S → A → B → D → G = 1 + 3 + 3 + 3 = 10

S → G = 12

Optimal path: S → A → C → G

Minimum cost: 4

Conclusion:

UCS finds S → A → C → G because it has the lowest total path cost of 4.